2026 IEEE Big Data Cup · TrafficFlowBench

Gated residual learning, physics-aware post-processing and as-of-origin queue forecasting

Final report of team **Emre Cirak** (Kaggle: biohack44)

Emre Cirak · MSc student, Gazi University, Ankara, Türkiye

Code: `https://github.com/synthdataofficial44/trafficflowbench-2026` · November 2026

	Private (April)	Public (March)	Note
Selected final submission	**0.88253**	0.87514	rank **55 of 802** on the private board, 78 places above the public rank
Task 1 + 3 (state, physics)	—	—	S_state ≈ 0.94 (noise floor ≈ 0.954), S_FD 0.988, S_LWR 0.61
Task 2 (queues)	S_queue ≈ 0.81–0.83	S_queue ≈ 0.77–0.79	probes and probe arithmetic (Sections 2.3, 6.1)
Task 4 (ODME)	S_ODME 0.994	S_ODME 0.994	exact projection; essentially closed

Abstract. We describe the pipeline behind our selected TrafficFlowBench submission and what the two leaderboards taught us about it. **Task 1** (state reconstruction, which also carries Task 3) is solved by five pooled LightGBM residual model sets that learn speed, per-lane flow and per-lane density as corrections of a space–time interpolation, trained on the training months and, transductively, on observed cells of the test months that are hidden and re-predicted; their blend is chosen on held-out test-month cells, and a two-stage physics lab (a density model and speed/flow heads that see conservation estimates) moves the answer towards vehicle conservation. **Task 2** (queue forecasting) replicates the organizers' window selector on the training months, builds features from the 60-minute history, a cell-transmission forecast and a month memory of recurring queues, and fits per-link onset and per-(link, step) ongoing classifiers in a full and a location-free variant; April models are refitted on training data plus March pseudo windows, and every window is decoded by an expected-IoU rule. Only released rows stamped at or before a window's origin feed its inputs, fitting and selection; a check that rebuilds the inputs with every later row removed finds them identical. **Task 4** is the exact Euclidean projection of the released prior onto the link counts (S_ODME ≈ 0.994). Every change passed a pre-registered gate on data the models never saw. Comparing all 46 submissions on both boards, Task 1 gains transferred to April one for one, the April refit was worth 0.005–0.011 of the total score, and month-specific Task 2 changes that won on March did not carry over to April.

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1 Overview of the solution

TrafficFlowBench scores four tasks on one composite,

$$S_{total} = 0.35 \cdot S_{state} + 0.30 \cdot S_{queue} + 0.15 \cdot S_{physics} + 0.20 \cdot S_{ODME},$$

computed per directional panel, averaged over the two directions of a corridor and then over five corridors. The data are synthetic but calibrated on real freeway detectors: ten directional corridors at 5-minute resolution, nine training months with answers, March 2031 for the public board and April 2031 for the private board, generated independently (five incidents in March, seven in April). Figure 1 shows the pipeline; each lane is one task.

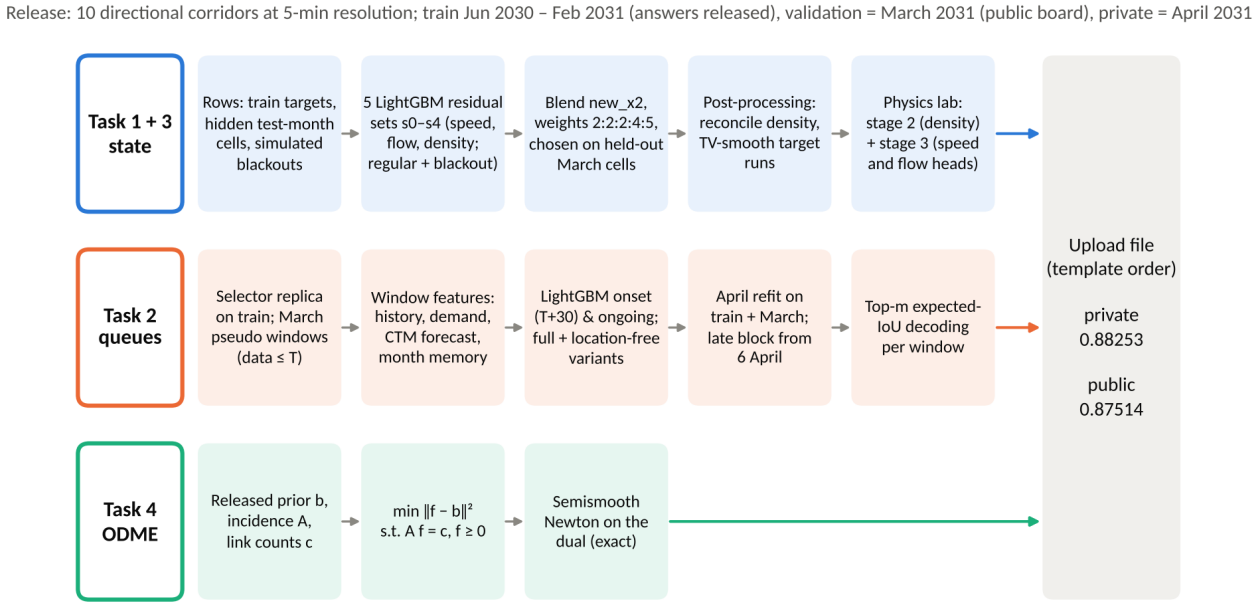


Figure 1. The pipeline of the selected submission. Task 3 has no rows of its own: the evaluator reads the Task 1 speeds and flows, so the physics work sits in the Task 1 lane.

The submission is the product of fifteen versions, each adding one step behind its own gate. The ones that survived, in order of their effect on the score:

- **Task 2 April refit** (v4–v15): the April models learn from the training months plus pseudo windows found in the March masked view; worth +0.005 to +0.011 of S_{total} on the private board (Section 6).
- **Task 1 residual sets and blend** (v2–v15): five LightGBM sets; the transductive rows from the test months are what makes Task 1 offline-learnable.
- **Physics post-processing and lab** (v2–v11): density reconciliation and total-variation smoothing, then a stage-2 density model and stage-3 speed/flow heads; with the larger blends, +0.025 S_{LWR} on the train holdout.
- **Task 4 projection** (v2): S_{ODME} from the NNLS baseline's 0.836 to 0.994.
- **Task 2 decoding and location-free variants** (v2–v13.1): top-m expected-IoU sets; location-light weighting of the onset model for a month whose bottlenecks differ from training.

2 Data, scoring and validation protocol

2.1 What is scored

Task 1 asks for speed and flow at blanked cells under three masking regimes (20, 30 and 50 % of eligible cells; each day belongs to one regime). $S_{state} = 0.54 \cdot S_v + 0.46 \cdot S_q$ per regime with $S_v = 1 - RMSE_{speed}/25$ km/h and $S_q = 1 - RMSE_{flow\ per\ lane}/600$ veh/h. **Task 3** scores the same numbers: $S_{physics} = S_{FD}/3 + 2 \cdot S_{LWR}/3$, where S_{FD} checks flow against a triangular fundamental diagram at the submitted density and S_{LWR} is an L1 norm of the violation of vehicle conservation, $N(t+\Delta t) - N(t) = \Delta t\,(q_{in} + q_{on} - q_{out} - q_{off})$, against the organizers' boundary flows.

Task 2 gives 60 minutes of history before an origin T and scores the queue state (speed ≤ $0.6 \cdot v_f$ in the noise-free state) at T+5 … T+30 by space–time IoU; 10 windows per corridor and month, half *onset* (no queue in the history)

and half *ongoing*; two corridors have none, so each split has 80 windows and each window is worth 0.00375 of S_{total}. **Task 4** asks for path flows f given the incidence matrix A, released link counts c and a weak prior b; S_{ODME} mixes OD agreement, link-count consistency, deviation from the prior and destination attraction. A cell counts only when pct_observed ≥ 75 (about 76 % of cells).

2.2 What can be measured offline

Only part of the score can be computed from released data, so each decision used the evidence closest to the scored quantity, and never the April labels:

- **Train holdout** (Task 1/3): every target of the last 30 training days with realistic blackouts. It gives S_{state} exactly and a conservation proxy S_{LWR} whose boundary flux is projected onto the released observations; the lab objective is $J = 0.35 \cdot S_{state} + 0.10 \cdot S_{LWR}$, the Task 1 + 3 part of the score.
- **Held-out test-month cells** (Task 1): 20,000 observed, eligible cells per panel and test month that no model trained on, hidden and re-predicted like targets. Task 1 is offline, so these cells measure the scored months themselves; March decides, April is only reported.
- **Selector replica and March pseudo windows** (Task 2): the official window selector (coverage ≥ 0.7, condition from the visible history, a queue in the horizon, persistence IoU ≤ 0.9, chronological greedy, five per condition, 360 minutes apart) replicated on the training months for training and cross-validation, and ≈ 2,450 ongoing / 234 onset pseudo windows found in the March masked view, labelled by its observed cells. March precedes April, so March windows may shape April rows; they never shape March rows (Section 4.6).
- **Task 4**: the link-count term and the deviation from the prior are computable; the reference path flows are not.

2.3 Reading the leaderboard by task

Three probe uploads built from one set of files read every component exactly: the files with every queue_pred at 0 give S_{queue}; every Task 1 value at 0 removes S_{state} and $S_{physics}$; every speed and flow multiplied by three leaves the per-cell density, hence S_{LWR}, unchanged while S_{state} falls to zero. For the v8 main file on the public board this gives S_{state} 0.9378, S_{FD} 0.9883, S_{LWR} 0.6084 ($S_{physics}$ 0.7350), S_{queue} 0.7734 and S_{ODME} 0.9939; zeroing the onset windows alone splits S_{queue} into onset 0.703 and ongoing 0.844. Figure 4 turns these into the points lost against a perfect answer. Because the probes change March and April rows alike, the private board reads them too: on April the v2 file had S_{queue} 0.807 (onset 0.720, ongoing 0.895) and the v8 file 0.827, with S_{ODME} 0.994 again.

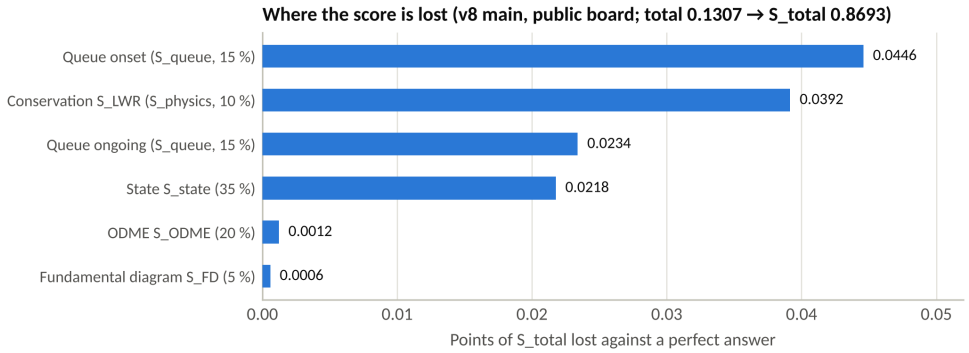


Figure 4. Points of S_{total} lost per component for the v8 main file (public board, exact probe arithmetic). Queue onsets and vehicle conservation dominate; the state RMSE comes next and is already close to the measurement-noise floor (Section 3.5).

2.4 Rules we held ourselves to

No external data, no pretrained weights. Task 1 is offline (organizer ruling), so observed cells of the test months are legitimate training rows. For Task 2, a window with origin T uses only released rows stamped ≤ T for its inputs, for model fitting and for model or threshold selection (rulings of 15 and 18 October); every March row stamped before an April origin may serve April windows, never the reverse; no April label is evaluated or used. Section 4.6 describes the check that enforces the input part of this rule.

3 Task 1 and Task 3: state reconstruction under physics

3.1 Representation and targets

Each panel is loaded into dense [time × link] arrays covering all three splits on one 5-minute axis, with links in physical order (index increasing downstream), so that upstream and downstream neighbours are index shifts. The model never predicts a level from scratch: for every target cell a space–time interpolation of the released neighbours is the baseline, and pooled LightGBM regressors (all panels together) learn the residual of speed, per-lane flow and per-lane density (q/v). Two families are trained: *regular* models for targets with same-time evidence, and *dark* models for cells inside the Task 2 blackout rows (horizon plus 60-minute buffer, every link blank), trained on simulated blackouts. Features describe the cell's neighbourhood in space and time, its distance to the nearest released cells, ramp flows and validity, and the link's fundamental-diagram parameters.

3.2 Training rows and model sets

Training rows come from three sources: masked targets of the training months (truth from the unmasked layer), **transductive** rows from the test months (observed cells hidden in rounds the way the mask hides them, then re-predicted), and simulated blackouts. Five sets differ in their row samples; sets 2–4 draw larger transductive samples, train deterministically and never see the held-out test-month cells. Table 1 lists them with the early-stopping RMSE of their six models (LightGBM, 255 leaves, learning rate 0.06, feature fraction 0.6, bagging 0.7, $\lambda_2 = 2$; dark models 63 leaves, rate 0.05). Every regular model stopped at its round cap still improving, which motivated the larger later sets.

Set	Transductive rows (regular / blackout)	Rounds	speed	flow	density	speed	flow	density
			regular	regular	regular	blackout	blackout	blackout
s0 (v8)	150,000 / 50,000	2,500	1.556	30.78	0.552	3.657	48.02	1.739
s1 (v8)	150,000 / 50,000 (own row seed)	2,500	1.586	30.80	0.548	3.613	48.12	1.742
s2 (v8)	300,000 / 100,000	2,500	1.575	30.41	0.524	3.630	47.77	1.744
s3 (v13)	450,000 / 150,000	3,500	1.548	30.28	0.542	3.556	47.33	1.700
s4 (v14)	600,000 / 200,000	5,000	—	—	—	—	—	—

Table 1. The five LightGBM sets. RMSE on each set's early-stopping rows in natural units (speed km/h, flow veh/h/lane, density veh/km/lane); s4's per-model figures were not kept in the reports, its effect is measured in Table 2. Blackout cells cost about 2.3 times the speed error and 1.6 times the flow error of regular cells.

3.3 Blend and post-processing

Members are blended with fixed weights. A candidate blend replaces the current one only if it gains more than two bootstrap standard errors of S_{state} on the held-out *March* cells (days resampled) and does not lose J on the train holdout by more than $2{\cdot}10^{-4}$. The final blend, *new_x2*, weights the sets 2 : 2 : 2 : 4 : 5 (fifteenths): each newer set at twice the average weight. It gained 0.00041 S_{state} on held-out March cells (≈ 10 SE) and 0.00038 on held-out April cells, which were never used to decide.

Two post-processing steps target Task 3. In dense traffic ($v < 0.6{\cdot}v_f$) the density implied by the speed and flow models is reconciled with the dedicated density model, and the change is split between speed and flow (speed share 0.75). Then, inside every run of consecutive target cells of a link, the density is smoothed by a total-variation step with the released cells at both ends fixed. The reason is the form of S_{LWR}: inside a run a..b it adds $|e_a| + \Sigma|e_{t+1} - e_t| + |e_b|$, the total variation of the density error padded with zeros. Independent cell-by-cell errors pay at every step and an isolated target pays $2|e|$, while S_{state} sees only the squared errors. The smoothing weights (τ by traffic regime) were chosen on the train holdout: raised S_{LWR} by +0.015 for a S_{state} cost of 0.0002 (Table 2).

3.4 The physics lab: stage 2 and stage 3

The lab works on the train holdout and only accepts a candidate whose J gain exceeds two day-bootstrap SEs in both halves of the holdout, does not lower S_{state} by more than $5 \cdot 10^{-4}$, and does not lose on the held-out March cells. Smoothing each link's log density along time (a Kalman filter with an RTS smoother) and pulling the free-flow density towards the density model were tested first and lost: the missing information is in the released cells around a target at the *same* time. **Stage 2** is a LightGBM model of the log-density error of a target cell from the answer and the released record around it (±1–3 slots, neighbouring links, the neighbours' answers, which the stage-1 models never see), cross-fitted by blocks of holdout days and applied with shrink 1.0. **Stage 3** adds what stage 2 does not see: each neighbour's deviation from its own previous and next slot (a fluctuation carried from cell to cell by the vehicles), the median flow of the released links of the same ramp-free segment (a conserved quantity in free flow), the link's ramp flows and two conservation estimates of the link's flow (from the upstream link and the ramps between, and from the downstream link and its ramps). Its heads predict log speed and log flow with an L2 loss (S_{state} is an RMSE) and log density with an L1 loss weighted by vehicle count (S_{LWR} is an L1 norm). Of three candidates the speed-and-flow heads (*s3vq*) won. Every blend change reruns the whole lab on the new answer.

Answer	Members / step	S_state	S_LWR	J	March cells
v8 raw blend	s0–s2, equal weights	0.94316	0.62056	0.39216	—
v8 answer	+ reconciliation + TV smoothing	0.94294	0.63545	0.39357	—
v11 answer	+ stage 2 + stage 3 (s3vq)	0.94404	0.64345	0.39476	0.94724
v14 answer	blend with s3 (new_x2), lab rerun	0.94428	0.64463	0.39496	0.94767
v15 B0	blend with s4 (new_x2) + post-processing	0.94351	0.63878	0.39411	—
v15 B1	+ stage 2	0.94360	0.64484	0.39474	0.94713
v15 answer (final)	**+ stage 3 (s3vq)**	**0.94446**	**0.64547**	**0.39511**	**0.94803**

Table 2. Task 1 answers on the train holdout (S_state exact, S_LWR proxy, J = 0.35·S_state + 0.10·S_LWR) and S_state on the held-out observed March cells (195,016 cells). Stage 2 adds +0.00064 J (SE 0.00003), stage 3 a further +0.00037 (SE 0.00003) and +0.00090 S_state on March cells (SE 0.00019). Public-board reads of each adopted change: v10 +0.00066, v11 +0.00053, v14 +0.00034, v15 +0.00022.

3.5 Error analysis

- **Close to the noise floor.** The scored value is the noisy observation itself. Bounding its noise from the training truth in free flow (second differences in time for speed; two neighbouring detectors without a ramp between them for flow) gives the S_{state} an answer would score if measurement noise were its only error: ≈ 0.954. Our holdout S_{state} is 0.944 and the held-out March cells read 0.948, so at most ≈ 0.008 S_{state} (≈ 0.003 of S_{total}) is left in Task 1.
- **Where the error sits.** On the models' early-stopping rows, blackout cells (whole rows blank around the Task 2 horizons) carry 3.6 km/h speed and 48 veh/h/lane flow RMSE against 1.55 km/h and 30.4 veh/h/lane for regular cells. The lab's breakdown places most of the S_{LWR} loss at isolated targets in free flow and in the long target runs of the 50 % regime days.
- **The fundamental diagram is not the issue.** S_{FD} = 0.988 without any constraint: separate speed and flow models that learn from detector data already produce FD-consistent pairs.
- **Conservation is, and the proxy tracks it.** S_{LWR} (0.61 on the public board for v8) is the largest physics loss. The train-holdout objective J predicted the board: stage 2 gained 0.00064–0.00067 J in the v14 and v15 labs, and the v10 file that introduced it read +0.00066 on the public board. Like S_{state}, S_{LWR} is bounded by measurement noise, since conservation is checked against noisy observations.

4 Task 2: online queue forecasting

4.1 Framing the two conditions

The two conditions are different problems. For **onset** windows the selector takes the earliest origin whose horizon touches a new queue (organizer-confirmed and visible in every training window), so the truth sits at T+30 only: onset is a per-link classifier for the last step, with steps 1–5 left empty. For **ongoing** windows a queue is visible at T and the question is how it spreads, shrinks or dissolves: a per-(link, step) classifier on the cells near queue activity. Training labels are observed speed $\leq 0.6 \cdot v_f$, with missing cells filled in time and, near observed queues, by a space–time imputer (the first queued slot of a new queue is sharp in time and smooth in space).

4.2 Features

History (v2): the 60-minute window and a 3-hour masked-view context per link and neighbour, speed and flow relative to v_f and capacity, the distance to the nearest queue, ramp flows and validity, demand build-up against the time-of-day profile, and the same time of day on the last seven days. **Physics** (p): a cell-transmission model (Daganzo, 1994) run forward from the state at T with three demand scenarios (flat, trend, time-of-day profile); its predicted queue extent at each step is a feature. **Month memory** (m): where queues sat at this time of day on the earlier days of the same month (from 30 minutes before T's time of day to 215 minutes after, read from the released masked view up to T). Location-identifying columns (link position, panel, train-month queue priors) can be dropped to give a *location-free* variant.

4.3 Models, adaptation and decoding

LightGBM binary classifiers: onset 63 leaves, 500 rounds, seven seeds; ongoing 127 leaves, 700 rounds, three seeds; final recipes v2+pm (onset) and v2+m (ongoing), each the average of the full and the location-free variant. The March rows use these *standard* models trained on the training months only. April rows come from an **April refit** on the training months plus every March pseudo window (weight 2), adopted for both conditions because the same refit, trained on the first half of March, beat the standard models on the second half (Figure 5); the onset refit blends its full and location-free versions. Ten April ongoing windows from 6 April on use a **late-block** refit that also learns from April's own pseudo windows of the earlier days (all stamped before those origins). Decoding takes, per window, the top-m cells that maximise the expected-IoU surrogate $\Sigma_S p / (|S| + \Sigma_{\text{not } S} p)$ of the window's own probabilities, so no threshold is fitted. In v13.1 the onset location-free variant got weight 0.75 (*nl75*), adopted under a cross-draw rule: a gain above two SEs on the March draw, both March reads positive, no loss on the official-style or first-week windows, three of four corridor families.

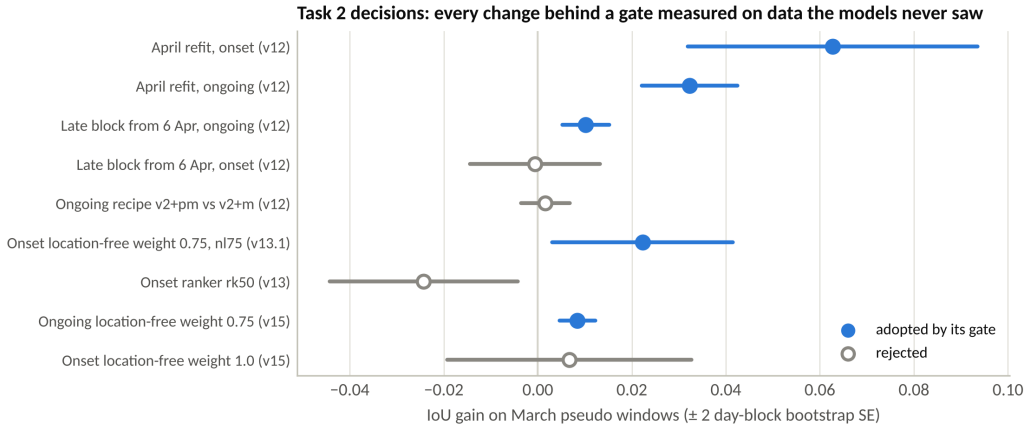


Figure 5. Gate statistics of the Task 2 decisions, measured on March pseudo windows (mean of the official-style and all-window IoU differences; the April-refit and late-block rows use the official-style windows of the second half of March). The ongoing weight 0.75 passed its gate in v15 but is not in the final file (Section 6.3).

4.4 Results and diagnostics

On the March pseudo windows the final onset recipe scores IoU 0.810 on official-style windows (0.786 before nl75) and the ongoing recipe 0.868 (0.879 on all 2453 windows). On the public board the onset part read 0.703 and the ongoing part 0.844 (v8 era); on April the v2 file read 0.720 and 0.895. Figure 6 shows where the errors are. Onset windows whose origin falls between 10:00 and 15:00 score 0.77 against 0.87–0.91 for the more regular early-morning onsets. The onset model finds the right site in 89 % of windows and then scores 0.92; the misses are not near-misses: the true queue lies a median 17 km away and is among the model's top five sites in only 42 % of them, consistent with incident-driven queues away from recurrent bottlenecks. For ongoing windows the difficulty is size: windows with one to five queued cells score 0.52, because one wrong boundary cell costs a large share of a small union, while queues of 25 or more cells score 0.92.

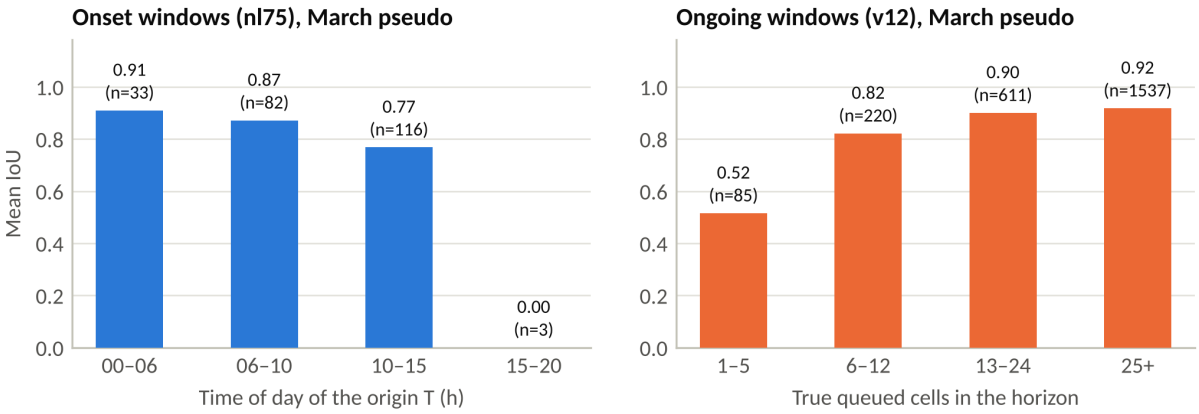


Figure 6. Task 2 diagnostics on March pseudo windows: onset IoU by time of day of the origin (left) and ongoing IoU by the size of the true queue in the horizon (right).

4.5 What did not work for Task 2

- **Eligibility-aware decoding** (v12): expected official IoU by Monte Carlo under the model's marginals and an eligibility rate, choosing m per window. CV −0.0015 (SE 0.0013); not adopted.
- **An onset ranker** (v13): a learning-to-rank LightGBM (rank_xendcg) whose scores are turned into probabilities by a temperature-calibrated softmax. −0.024 on March pseudo windows, −0.0006 on the public board.
- **Availability and empirical-capacity features** (v11) and a **CTM ongoing recipe** (v12): no gain beyond their gates.
- **A causal CNN** over the space–time history (v16) and a **CatBoost queue trial** (v18): public 0.85682 and 0.87187 against 0.8748–0.8749 for the files they were built on; neither changed the picks.

4.6 As-of-origin compliance

The input rule is checked by rebuilding the Task 2 inputs of official windows from a copy of the panel arrays in which **every released row stamped after the window's origin is blank**: all mainline channels and quality flags, ramp flows and ramp validity counts, with the month-memory, recency and CTM caches rebuilt from the truncated copy. One onset and one ongoing window per panel and month (32 windows) were compared column by column with the inputs built from the full arrays: all 141 onset and 118 ongoing model columns, and every other column, were equal. The check was validated by planting features that read the mainline two hours after T, on-ramps inside the horizon, or today's memory past T; each was flagged. Equal inputs and frozen models give unchanged predictions. Fitting and selection follow the same rule by construction (Table 3). One violation was found and repaired: in v13.1 the nl75 weighting, chosen on March pseudo windows that lie after the scored March origins, had been written into the March onset rows; v15 put those rows back to the standard models', and the final file carries them.

Model role	Fitted on
Standard models (March rows; hedge April rows)	train split only (selector-replica windows, train labels)
April refit (main April rows)	train + every March window (pseudo windows from the March masked view); every March row is stamped before every April origin
Late block from 6 April (10 ongoing windows)	train + March + April pseudo windows of the days before 6 April (labels read up to the block start)
Gates and selection (cross-draw rule, nl75, v15 gates)	March pseudo windows - they shape April rows only
Decoder	top-m expected-IoU surrogate of the window's own probabilities (no fitted threshold)

Table 3. What each Task 2 model role was fitted on (from the manifest of the selected submission). No April label is ever read.

5 Task 4: OD matrix estimation as an exact projection

For each panel and split we return the Euclidean projection of the released prior onto the released link counts, $f = \arg\min \|f - b\|^2$ subject to $A_{meas}\, f = c$ and $f \geq 0$. The leaderboard showed why it works so well: the scores are consistent with reference path flows produced by a ridge fit of the split prior to hidden targets, with the counts published as $A\, f^*$, in which case the KKT conditions of the projection coincide with those of the generator. The problem is solved exactly by a semismooth Newton method (Qi and Sun, 1993) on the dual over the distinct count rows (screenlines; at most ≈ 100 unknowns per split). $S_{ODME} = 0.9939$ on March and 0.994 on April against 0.836 for the NNLS baseline; the remaining 0.0012 of S_{total} was not worth pursuing.

6 Two months, two boards: what transferred to April

The private board scores April, a month generated independently of March with more incidents (seven against five) and scored Task 2 windows that fall mostly on weekdays. We uploaded 46 files, many of them probes that change one component, so the two boards can be compared component by component (Figures 2 and 3, Table 4).

Change (files compared)	Δ public	Δ private	Reading
Task 1 v10 → v11 (v11 main vs its v10-Task-1 probe)	+0.00053	+0.00052	transfers 1 : 1
Task 1 v11 → v14 (v14 main vs v13.1 main)	+0.00034	+0.00029	transfers
Task 1 v14 → v15 (v15b main vs v15 main)	+0.00022	+0.00019	transfers
April refit, v7 (main vs hedge)	+0.00099	+0.01061	largest April gain
April refit, v8 (main vs hedge)	+0.00100	+0.00979	
April refit, v15b (main vs hedge)	+0.00000	+0.00677	March rows identical
Month memory + refits, v8 → v10 (Task 1 also +0.0007)	+0.00380	-0.00331	wins March, loses April
Onset nl75, v12 → v13.1 (main)	-0.00008	+0.00041	mixed
Onset nl75, v12 → v13.1 (hedge)	-0.00008	-0.00064	mixed
Ongoing weight 0.75 on March rows (v15 probe vs main)	-0.00196	+0.00000	public read only
Ongoing weight 0.75 on April rows (v15 hedge vs hedge0)	+0.00000	+0.00062	April sides with the gate
Causal CNN for Task 2 (v16 vs v14 main)	-0.01802	-0.01678	bad on both

Table 4. Pairs of uploads that differ in one component. Task 1 changes move both boards alike; Task 2 changes chosen on March do not.

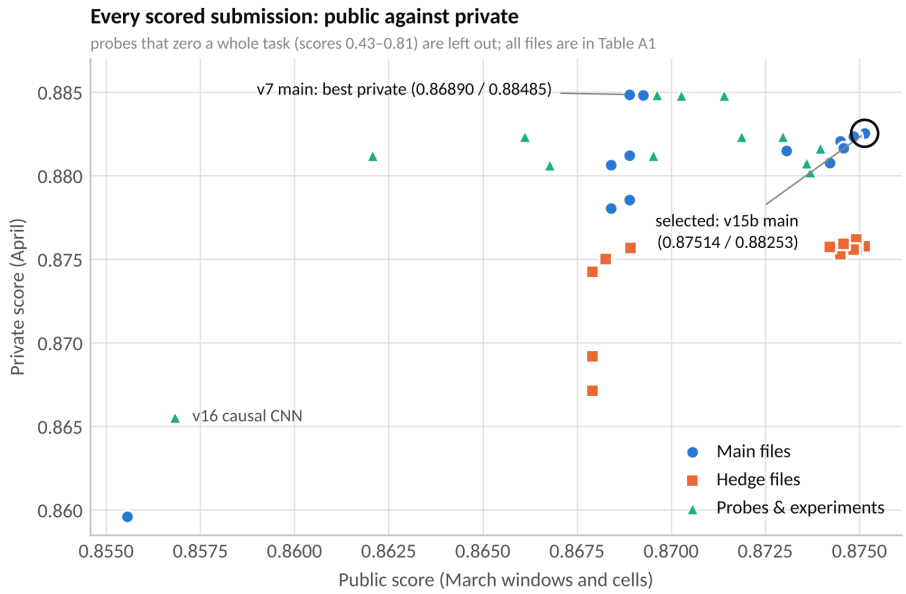


Figure 2. Public against private score for every scored upload. Hedge files sit lower on April: they lack the April refit. Probes that change only March rows share their base's private score (horizontal rows of triangles).

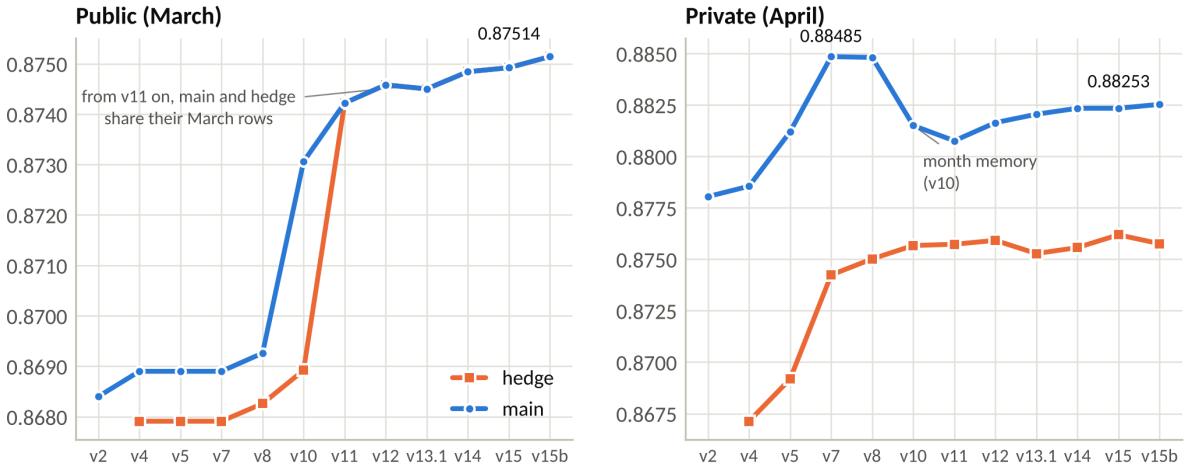


Figure 3. Main and hedge files by version on both boards. The public score rose at every step; the private score peaked with v7–v8 and dropped when the month memory entered the April rows (v10), then recovered slowly.

6.1 Findings

- **Task 1 gains transfer one for one.** Every Task 1 change read the same on both boards to within 0.00005. Held-out cells of the test months are the scored distribution itself, so Task 1 could be developed safely.
- **Adapting to the scored month mattered most.** The April refit on training data plus March pseudo windows was worth +0.005 to +0.011 of S_{total} on April (main against hedge, v7–v15b), i.e. +0.017 to +0.035 of S_{queue}, larger than any other single change of the competition.
- **Month-specific Task 2 recipes did not transfer.** The month memory and the refits built on it (v10) gained +0.0038 on March and lost −0.0033 on April. From the probe arithmetic, the Task 2 changes after v8 raised S_{queue} on March by ≈ +0.014 and lowered it on April by ≈ −0.013; our best April file was v7 (0.88485, about 40th place), not the selected one. A plausible reading: a memory of where queues sat earlier in the month helps when incidents recur in place and misleads when they move, and April had more incidents.
- **Small public samples mislead; large internal ones less so.** The ongoing weighting gained +0.0084 (4.5 SE) on 2,453 March pseudo windows but read −0.00196 on the 40 scored March ongoing windows — about half a window. Our pre-registered rule then switched to the fallback pair. On April the change was worth +0.0006 in the hedge file, siding with the pseudo windows. (The main file was unaffected: its April ongoing rows came from the refit, which kept the old weights.)

6.2 The leaderboard in context

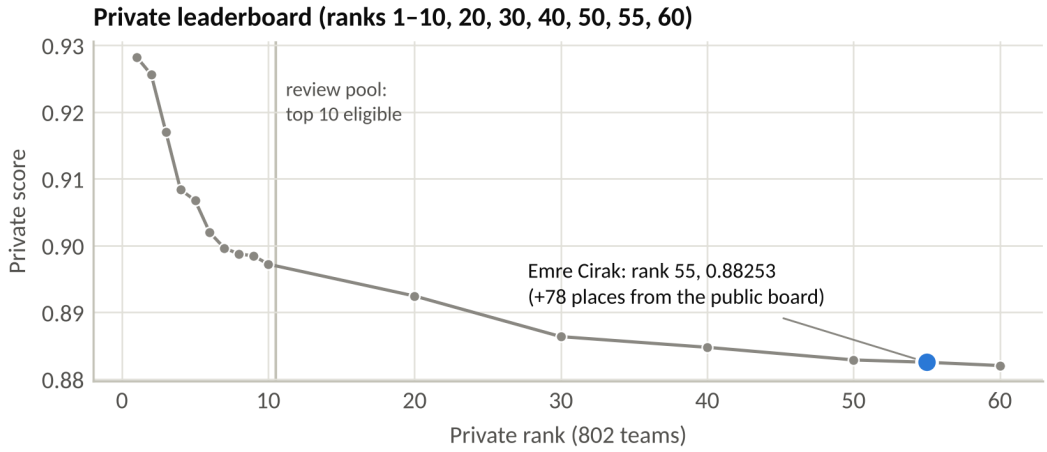


Figure 7. The private board around our rank (scores at the listed ranks only). The award review pool is the top ten eligible teams (private 0.89721 at rank 10).

We finished 55th of 802 teams with 0.88253, 78 places above our public rank, which suggests the method was less tuned to March than many. The gap to the top ten (0.89721) is 0.0147. With at most ≈ 0.003 left in Task 1 (Section 3.5) and ≈ 0.001 in Task 4, closing it would have needed about +0.04 of S_{queue}: better queue forecasts, above all for onsets.

6.3 How the final files were chosen

Each version produced a *main* file (bet: April resembles March, so learn from March) and a *hedge* (bet: April resembles the training months), with identical March rows from v11 on, so the two picks differed only in the April refit. Picks followed rules fixed before the reads (revert if a change reads ≤ −0.001 on the public board; switch to the fallback pair if the ongoing probe reads ≤ −0.0015). No final selection was made by hand, so Kaggle selected the two best public files, v15b main and v15b hedge (both 0.87514); the main file counted. Its Task 2 rows equal both v15 main and the fallback main (same file, md5 9be1bef8…), so the fallback rule changed only the hedge.

7 Negative results

Idea (version)	Measured	Outcome
Kalman / RTS smoothing of log density along time, 'rts' (v10 lab)	lost J on the train holdout	rejected
Free-flow density pulled towards the density model, 'kfree' (v10 lab)	below stage 2 on J	not adopted
Density head s3k, all heads s3all (v11 lab)	below s3vq on J	not adopted
CatBoost member of the Task 1 blend (v15, A100)	−0.00005 to −0.00075 on held-out March cells	rejected
Task 1 CNN / residual learners (v19, v21)	own confirmation gates	rejected
Eligibility-aware onset decoding (v12)	CV −0.0015 (SE 0.0013)	not adopted
Onset ranker rk50 (v13)	−0.024 March pseudo; −0.0006 public	rejected
Availability / empirical-capacity features (v11)	below their two-draw gate	not adopted
Ongoing recipe with CTM forecast v2+pm (v12)	+0.0016 (SE 0.0026)	not adopted
Onset location-free weight 1.0 (v15)	+0.0067 (SE 0.013); 1 of 4 families	not adopted
Causal CNN for Task 2 (v16)	public 0.85682 (−0.018)	never a pick
CatBoost queue blend (v18)	public 0.87187 (−0.003)	never a pick

Table 5. Ideas that did not make the final file, with the measurement that stopped them.

8 Reproducibility

The code is at `https://github.com/synthdataofficial44/trafficflowbench-2026` under the MIT license. Its package `tfb/` (24 modules) is byte-identical to the code that produced the selected file: `run_all.py` compares every module's SHA-256 with the manifest written by that run. The single entry script runs the stages of the original runs in their order with their settings (`configs/final.json`): caches, Task 4, the Task 1 chain (v8 sets → v10 lab → v11 lab → v13 set → v14 blend → v14 set → v15 blend and lab rerun), the Task 2 chain (v12 → v13.1 → v15), the upload file with its md5 against the selected one, and the as-of-origin check. Each stage stores its result with the code hash, the settings and a fingerprint of the data and is skipped when it finds one, so a run resumes and, given the trained artifacts, reproduces the file bit for bit.

Item	Value
Command	python run_all.py --data <unpacked competition data>
Hardware used	Google Colab Pro+, A100 runtime with High-RAM; CPU LightGBM only (no GPU needed)
Software	Python 3.13.16, numpy 2.1.3, pandas 2.2.3, pyarrow 23.0.1, lightgbm 4.6.0, scipy 1.16.3
Run time (full retrain)	Task 1 ≈ 20–26 h, Task 2 ≈ 5 h, the rest ≈ 40 min; with trained artifacts ≈ 1 h
Selected file	submission_20261025_1742_v15b_main.zip, submission.csv md5 6d196f7870a9e58d13edb4c183cef020
Components (md5)	task1_v15.parquet c0327df4…, task2_v15_0.csv 9be1bef8…, task4.csv d514ac20…
Decisions pinned	the Task 2 file (fallback main, a public-board read); everything else is decided by the code's own gates
Determinism	sets s2–s4 train with deterministic=True; other fits are multithreaded CPU LightGBM and may differ in the last digits across machines, far inside the 1 % tolerance (±0.0088)

Table 6. Reproducing the selected submission. The repository README gives the details, including a control-flow test of the entry script that needs no data.

9 Limitations and what we would do next

- **Onset forecasting of incident queues.** Most remaining points are onset windows whose queue appears where no recurrent bottleneck exists. History alone cannot announce an incident; a better use of the evidence would be detector-level precursors (speed variance, occupancy rises) within the 60-minute window, and a calibrated abstention when no site is likely.
- **Choosing on the right draw.** April showed that changes learned from one month's demand draw generalise poorly. A selection rule averaging the training-month CV and March pseudo windows, rather than letting March decide alone, would have kept v7–v8's April rows.
- **Conservation as a constraint.** S_{LWR} was improved by learned corrections; projecting the density field onto the conservation equations with the released ramp flows, as Task 4 projects path flows onto counts, is the natural next step.
- **Run time.** A full retrain takes about a day; the incremental, gated design made it safe to change one thing at a time but costly to rebuild.

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TrafficFlowBench organizers (2026). TrafficFlowBench public repository: loaders, baselines and evaluators. github.com/jacky850/trafficflowbench-public.

Appendix A Every upload with both scores

All 46 scored uploads, newest first (descriptions shortened from the upload messages). Highlighted: the file that counts for the final ranking. Probes that change March rows only share their base file's private score.

File	Content	Public	Private
20261025_1742_v15b_hedge	tfb v15b hedge: T1 v15 (new_x2) | T2 v15 fallback hedge | T4 L2-proj	0.87514	0.87576
20261025_1742_v15b_main	**tfb v15b main: T1 v15 (new_x2) | T2 v15 fallback main | T4 L2-proj**	**0.87514**	**0.88253**
20261020_2242_v15_hedge0	tfb v15 fallback hedge: T1 v14 new_x2 | T2 v13.1 hedge April rows, March = v12 | T4 L2-proj	0.87492	0.87557
v18_queue_5813ff11f75e	tfb v18 queue trial: v15 + train-only CatBoost blend (experimental)	0.87187	0.88234
v17_diagnostic_onset_ab385968c0f5	tfb v17 diagnostic: v15 main with March onset from v16 (April unchanged)	0.86612	0.88234
v16_37990620aa26	tfb v16 experimental: v14 T1 + causal CNN/tree T2 + L2 T4	0.85682	0.86556
20261020_2242_v15_probe	tfb v15 probe_og: v15 main with March ongoing rows at w0.75	0.87296	0.88234
20261020_2242_v15_hedge	tfb v15 hedge: T1 v14 new_x2 | T2 v15 hedge (train-only + memory; April ongoing w0.75) | T4 L2-proj	0.87492	0.87619
20261020_2242_v15_main	tfb v15 main: T1 v14 new_x2 | T2 v15 main | T4 L2-proj	0.87492	0.88234
20261020_1913_v14_hedge	tfb v14 hedge: T1 v14 (new_x2) | T2 v13.1 hedge | T4 L2-proj	0.87484	0.87557
20261020_1913_v14_main	tfb v14 main: T1 v14 (new_x2; v10 s2 + v11 s3 refit) | T2 v13.1 main (onset nl75) | T4 L2-proj	0.87484	0.88234
20261014_2114_v131_hedge	tfb v13.1 hedge: T2 train-only + month memory, onset nl75 | T1 v11 | T4 L2-proj	0.87450	0.87528
20261014_2114_v131_main	tfb v13.1 main: T2 onset nl75 + v12 ongoing | T1 v11 | T4 L2-proj	0.87450	0.88205
20261014_1350_v13_probe_rk50	tfb v13 probe_rk50: v12 main with the onset ranker on March onset rows	0.87396	0.88164
20261014_0905_v12_hedge	tfb v12 hedge: T2 train-only + month memory, onset top-m decoder | T1 v11 | T4 L2-proj	0.87458	0.87592
20261014_0905_v12_main	tfb v12 main: T2 onset top-m decoder, ongoing v2+m x3 seeds | T1 v11 | T4 L2-proj	0.87458	0.88164
20261013_0837_v11_hedge	tfb v11 hedge: T2 onset v2+pm / ongoing v2+m train-only + month memory | T1 v11 s3vq | T4 L2-proj	0.87422	0.87573
20261013_0837_v11_main_t1base	tfb v11 main_t1base probe: v11 main with the v10 Task 1	0.87369	0.88024
20261013_0837_v11_onset_v10	tfb v11 onset_v10 probe: v11 main with the v10 March onset rows	0.87360	0.88076
20261013_0837_v11_size	tfb v11 size probe: every March onset window at its top cell	0.81183	0.88076
20261013_0837_v11_main	tfb v11 main: T2 onset v2+pm / ongoing v2+m | T1 v11 s3vq | T4 L2-proj	0.87422	0.88076
20261012_1639_v10_hedge	tfb v10 hedge: T2 v9 hedge (train only, no memory) | T1 v10 | T4 L2-proj	0.86892	0.87567
20261012_1639_v10_main	tfb v10 main: T2 month memory (onset v2+pm, ongoing v2+m) | T1 v10 B0+s2 | T4 L2-proj	0.87306	0.88150
20261012_1639_v10_onset_zero	tfb v10 onset_zero: v8 main with every March onset window 0	0.76385	0.88481
20261011_0318_probe9mo	tfb v9 probe9mo: v8 main + March ongoing rows from v2+m	0.87027	0.88481
20261011_0318_probe9m	tfb v9 probe9m: v8 main + March onset rows from v2+pm	0.87140	0.88481
20261011_0227_decomp_scaled	tfb v9 decomp scaled: v8 main with Task 1 speed and flow x3	0.52219	0.53743
20261011_0227_decomp_no_state	tfb v9 decomp no_state: v8 main with every Task 1 value 0	0.43078	0.44680
20261011_0227_decomp_no_queue	tfb v9 decomp no_queue: v8 main with every queue_pred 0	0.63725	0.63675
20261010_2236_final_hedge	tfb v8 hedge: T1 v8 blend | T2 train only | T4 L2-proj	0.86826	0.87502
20261010_2236_final	tfb v8 main: T1 v8 blend | T2 v2, April refit on train+March | T4 L2-proj	0.86926	0.88481
20261009_2122_final_probe7p	tfb v7 probe7p: main + March onset rows from probe5 recipe + CTM physics forecast	0.86962	0.88485
20261009_2122_final_hedge	tfb v7 hedge: T1 2 sets | T2 train only, late April refit | T4 L2-proj	0.86791	0.87424
20261009_2122_final	tfb v7 main: T1 2 sets | T2 v2, April refit on train+March, late April refit | T4 L2-proj	0.86890	0.88485
20261009_1603_final_probe6	tfb v6 probe6: main + March onset rows from the v6 onset recipe	0.86208	0.88120
20261009_1603_final_probe5	tfb v6 probe5: main + March onset rows from the v5 April onset recipe	0.86953	0.88120

File	Content	Public	Private
20261008_2236_final_hedge	tfb v5 hedge: T1 2 sets | T2 train only | T4 L2-proj	0.86791	0.86918
20261008_2236_final	tfb v5 main: T1 2 sets | T2 v2, April refit on train+March | T4 L2-proj	0.86890	0.88120
20261008_1422_final_hedge	tfb v4 hedge: T1 LGB+transductive+TV | T2 train only | T4 L2-proj	0.86791	0.86713
20261008_1422_final	tfb v4 main: T1 LGB+transductive+TV | T2 v2, April refit on train+March | T4 L2-proj	0.86890	0.87855
20261007_1359_t2probe	tfb v3 probe: v3 March rows	0.86677	0.88064
20261007_1359	tfb v3: T1 LGB+transductive+TV | T2 + current-month recency, online onset adaptation | T4 L2-proj	0.86840	0.88064
probe_onset_zero_20261007_0749	probe onset_zero of v2	0.76299	0.77009
probe_queue_zero_20261007_0749	probe queue_zero of v2	0.63639	0.63584
20261007_0045	tfb v2: T1 LGB+transductive+TV | T2 selector LGB + March-gated April adaptation | T4 L2-proj	0.86840	0.87805
submission.csv	first submission (baseline-style)	0.85556	0.85960

Table A1. Every upload with its public and private score.